

Routes are a Strategy because the Trip has one job, giving the next settlement, and there are several swappable ways to do it. Warmest, fastest and cheapest are whole algorithms for the same request. The player picks one and can change it whenever they like. Swapping the route doesn't change what the Trip is, only how the answer is worked out. That is Strategy: a family of interchangeable algorithms chosen by the client.

Movement is State because the Traveller changes its own behaviour as its condition changes. When gold drops below the cost of the current mode, the mode itself puts the next one in place. The player might pick a starting mode, but the transitions are never theirs to make. That is State: an object that alters its behaviour when its internal state changes.

The diagrams look the same because both hand work off to an abstract interface with three subclasses. The difference is who decides. In Strategy the client decides and any strategy is valid at any time, CheapestRoute doesn't even know FastestRoute exists. In State the states decide, and only certain states may follow others, TeleportationCircle knows about MerchantCaravan and creates it. Routes as State would mean a Route deciding when to turn into another Route, which makes no sense. Movement as Strategy would mean GameManager checking the gold and picking a mode, which puts the rules in the client, the very fault from Task 1.